

senses sound or vibration, it generates an electrical signal. This weak electrical signal is amplified and rectified and the resulting DC voltage is given to the switching circuit for further operation. Two NE555 timers are used—the first wired as a low frequency astable multivibrator and the second as an audio-frequency astable multivibrator.

The working of the circuit is simple. When you produce sound in the vicinity of the condenser microphone, the weak sound signals are amplified and switched by transistors (T1 through T6) to provide a

12V DC at pin 4 of IC1. This enables the LEDs (LED1 and LED2) to flash for a few seconds. Normally, pin 4 of IC1 (NE555) is held low by resistor R13 (470-ohm) to ground. But when the microphone senses sound or vibration, DC voltage is developed across resistor R13 (470-ohm) and pin 4 of IC1 goes high. At the same time, pin 4 of IC2 also becomes high. As a result, the speaker produces a barking sound. When the circuit senses sound through the microphone, the dog starts barking for a few seconds and the LEDs start flashing. By adjusting a 22k preset

(VR1) connected in the second timer, you get a barking sound. It is advisable to keep the microphone away from the speaker to avoid a feedback noise.

The circuit is assembled on a general-purpose PCB and housed in a suitable cabinet. The condenser microphone is connected with a pair of wire. LED1 and LED2 are connected by wires on the front panel or directly soldered on the PCB board for a visual indication. A 12V battery or a standard 12V, 250mA adaptor is used for powering the circuit. ●

DOOR-OPEN ALERT

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If you forget to close the door of an air-conditioned room, it will result in high electricity bill. The door-open alert described here beeps if you leave the door open for more than 30 seconds. When the door opens, the reed switch (normally of the closed-type) activates and the IC starts counting down.

The circuit is built around 4060 (IC1) and some discrete components. 4060 is a 14-stage counter/divider/oscillator. The frequencies can be decided through R3 and C2. The reed

switch S1 is an electrical switch operated by an applied magnetic field. It consists of a pair of contacts on ferrous reeds in a hermetically sealed glass envelope. There are two types of reed switches—normally open and normally closed. The normally open reed stays open when there is no magnet near the switch and closes when there is a magnet nearby. The normally closed reed is closed when it is not near a magnet and opens when it is near a magnet. In this circuit, the normally closed type reed switch is used. Fig. 1 shows the door-open alert circuit.

Whenever the door is opened, the reed switch closes and makes pin 12 of IC1 low and enables it to oscillate. After a pre-set delay of about 30 seconds, the piezoelectric buzzer beeps intermittently

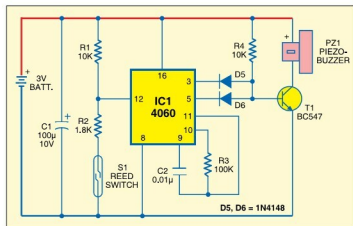


Fig. 1: Alert door opening circuit

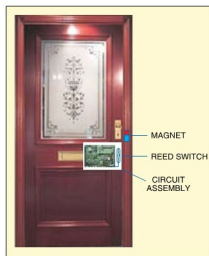


Fig. 2: Circuit assembly

for 30 seconds, then stops for the same period of time, and the cycle repeats until the door is closed. The circuit can run either from a 3V DC adaptor or from a couple of 1.5V battery cells. The circuit is assembled on a small piece of general-purpose PCB and a magnet is fixed on the door. The circuit assembly is shown in Fig. 2. ●